

Who Pays for the Energy Transition? Distributional Evidence and Policy Lessons for Switzerland

Policy white paper

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Executive Summary

Energy and climate policy instruments create direct and indirect costs and benefits for consumers and other parts of society in various ways. The distribution of these costs and benefits and how they are perceived are important determinants of public opinion on policy proposals. This was evidenced in the political discourse around the CO₂ Act in 2021 and is supported by findings from the most recent population survey conducted by the SWEET consortium EDGE. Therefore, it is important for policymakers to understand how the impacts of different policies are distributed across the population and across stakeholders. This knowledge allows for transparent information in the political process, which facilitates a level-headed political discourse about the choice of policy instruments. Furthermore, anticipating unfavorable distributional outcomes allows policymakers to refine policy design and implement complementary policies that mitigate negative impacts on parts of society that merit protection.

Several studies conducted by EDGE analyze distributional aspects around policy proposals in Swiss climate and energy policy and survey public opinion and perceptions. One strand of research focuses on the employment effects of ambitious renewable electricity targets. It finds that employment linked to constructing, operating and dismantling power plants would be larger under renewable electricity targets, but the impacts would differ across industries and regions. The industry sector, especially power plants and construction, would benefit more than the service sector. The regional distribution of employment impacts depends strongly on how the utilisation of electricity generation technologies changes between the scenarios, where additional power plants are located and how the regions participate in the supply chains of constructing, operating and dismantling power plants. Less prosperous cantons tend to enjoy larger employment impacts, which could slightly reduce inequalities. The employment impact is higher in the midlands and the Alps than in the cities.

A second strand of research analyzes how dynamic electricity tariffs, designed to increase the efficiency of the electricity system, impact different parts of the Swiss population. The findings show that the patterns in the distribution of impacts depend on how well households are equipped to respond to dynamic tariffs and on projected future electricity price patterns. As long as heat pumps, solar PV, batteries, and battery-electric vehicles are not widespread, dynamic tariffs can impose additional costs on a significant portion of the population. The distribution of impacts does not warrant serious concern about burdening certain population groups with disproportionate policy costs; rather, it shows that, across all subgroups of the Swiss population, the heterogeneity of impacts is large. This points to a fundamental issue with cost-efficient policies, which often require clear price signals for consumers and, through heterogeneity in spending patterns, produce wide variation in impacts even within distinct subgroups of the population. In the case of dynamic electricity tariffs, opt-out options (currently mandated by federal law) can help avoid negative impacts, and by making sure that efficient and flexible technologies are widespread, it can be ensured that the majority of the population can profit from dynamic tariffs and have an incentive to opt for them.

A third research focus of EDGE examines policies that mitigate emissions in the residential sector, quantifying the effectiveness and distributional effects of a range of policy mixes. These mixes combine price-based incentives (taxes and subsidies) with regulatory instruments (bans on fossil-fuelled boilers and obligations to retrofit

building envelopes and install PV), alongside protection measures to shield financially vulnerable households from disproportionate burdens. The studies find that, under a combination of heating bans and PV and retrofit obligations proposed in the MuKE 2025, as well as price-based incentives, it is less affluent households, older single-person households, homeowners, rural households, and households in less densely populated cantons that risk having to make the least favorable investments. Similar but more stringent policy mixes forcing investments by 2035 would accelerate defossilization, but may have unintended side effects such as liquidity constraints for some dwelling owners, increased pressure on public spending, workforce bottlenecks, reduced revenues for utilities, and increased energy-related costs for certain households. The implications of these findings are: first, ambitious time-bound bans and obligations should be phased in at a pace that matches the availability of skilled labor and capacity in the building sector, while both should be expanded to meet the demands. Second, particularly if ambitious bans and obligations are introduced, well-targeted support instruments (such as exemptions or subsidies) can help protect financially vulnerable groups without unnecessarily increasing public spending or undermining policy effectiveness. Third, policymakers should anticipate that utilities can face declining revenues and changing service demands and thus may need to adapt their business models. This may require changes in regulation.

The “SWEET EDGE survey” from 2025, finally, asked Swiss residents about their preferences regarding two regulations for mitigating emissions in the residential sector: the installation ban on fossil-fuelled heating and the retrofit obligation. The survey finds that public support for these policies decreased considerably when individuals were confronted with the policies’ distributional impacts on both homeowners and tenants. The results indicate that the public support of ambitious defossilization policies is likely to be reduced as soon as distributional impacts enter the public discussion. Moreover, the protection measures that were proposed – targeted exceptions and additional subsidies for financially vulnerable households – were unable to increase public support, likely because they did not create a clear perception of improved distributional impacts. Stronger support measures may be needed to compensate for the adverse effects on both vulnerable homeowners and tenants.

In summary, the following general recommendations can be made to policymakers.

1. Make sure that the Swiss economy is well prepared for the changes entailed by defossilization. Sufficiently skilled labor and appropriate business models are needed to avoid consumers remaining stuck with expensive options.
2. Ensure that all actors have the right incentives and information. Split incentives and inattention in the owner–tenant relationship may lead to investments in demand flexibility not being made that would be desirable for tenants.
3. Credit constraints can restrict consumers’ access to efficient and flexible technologies. Low-interest loans, “energy-as-a-service”, and “rent-to-buy” business models can help. The offers must be available to and known by the consumers.
4. Transparent and proactive communication is key. If policy impacts are heterogeneous, different political actors can pick different examples to promote their view of a policy. The resulting contradictory arguments can leave the impression that policy options are poorly understood or not well thought through.
5. Heterogeneity of impacts violates the equal sacrifice principle and may appear unattractive to the population and thus to voters. For policy proposals to succeed,

the benefits should be emphasized, and the arguments that the benefits outweigh the possible inequality in and uncertainty about the distribution of policy impacts should be convincing.

6. Where vulnerable parts of the population are particularly exposed to negative policy impacts, put complementary policy instruments such as exemptions or social welfare measures in place.

1 Introduction

Switzerland, under the Paris agreement, has a legal obligation to reduce its greenhouse gas (GHG) emissions, and surveys show that the Swiss population takes climate change seriously (see, e.g., results from EDGE's second survey, Fig. 1).

At the same time, poverty is another important issue on people's minds, and the political discourse in 2021 around the revision of the CO₂-act revealed that the distribution of its impacts across different parts of the population was a critical criterion for voters' views about the policy proposal.

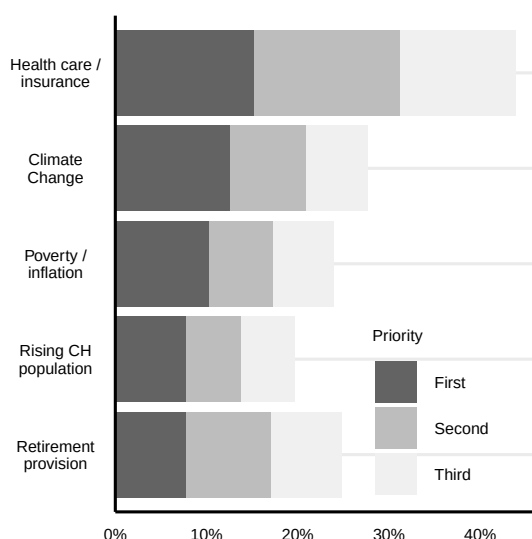


Figure 1: Top 5 answers from EDGE Survey 2 (May to August 2025, 1857 respondents) to the question “Which of these topics do you think are the three most important issues facing Switzerland?”

It is, therefore, important for political stakeholders to know beforehand what distributional effects of introducing different proposed policies will have. If distributional effects are known, policymakers can ensure that politicians and voters are well informed and that the discussions around equity remain level-headed. If policy impacts are perceived as distributed unfairly, alternative designs

or complementary policies to improve fairness should be considered to ensure public support. Policy packages that result in fair outcomes are better aligned with society's moral obligation to support its least well-off constituents and are more likely to succeed politically with voters.

We invite the reader to look at the distributional impacts described herein through two lenses. The first, frequently discussed way of assessing distributional effects compares the distribution of consumer welfare across income levels before and after a policy intervention. A policy that causes consumer welfare of the financially vulnerable to decrease by less than another policy instrument makes the former preferable under this assessment.

The second perspective does not require protection of the financially vulnerable but strives for equality in impacts (equal sacrifice principle): All households should experience similar (relative to their means) gains or losses from a policy intervention. While assessing policy instruments under this lens has some undesirable properties (both the introduction and a later retraction of a policy intervention can rank badly under this assessment), policy impacts on individuals are more salient than final outcomes for consumer welfare.

The SWEET consortium EDGE has carried out a set of studies that analyze how perceptions and policy impacts are distributed across different parts of Swiss society and the economy for a selection of policy instruments that promote the energy transition in Switzerland. This white paper provides an overview of their main findings and highlights the consequences for policy-making that they entail. It groups EDGE's studies into

four themes:

Employ “*Distribution of employment impacts of the transformation in the electricity sector*” (Section 2) – Input–output modeling is combined with techno-economic data to evaluate the employment impacts of transforming the electricity system according to scenario results by the EXPANSE model. The study compares employment effects across industries and regions (cantons).

Ele “*Distributional effects of promoting electricity demand flexibility through dynamic tariffs*” (Section 3) – This study uses a model of rational agents that optimize their buildings’ energy systems and energy use patterns to analyze how households in the buildings can profit or lose if they have to pay dynamic electricity tariffs rather than constant ones.

Res “*Distributional effects of the defossilization in the residential sector*” (Section 4) – This theme summarizes two microsimulation-based studies that consider different regulations and ambition levels to decarbonize the residential sector. Direct cost impacts on industry, utilities, the state, dwelling owners, and different household groups are analyzed.

PO&P “*Public opinion and perceptions of distributional effects in the residential sector*” (Section 5) – A survey-based experiment uses results from the theme **Res** and analyzes how policy acceptance of homeowners and tenants change when respondents are confronted with information about distributional impacts.

Figure 2 gives an overview of the themes and the dimensions across which they report distributional impacts. The work carried out by EDGE cannot cover the distributional impacts of all policy options needed for a successful implementation of Switzerland’s energy transition, nor can it cover all dimensions across which policy impacts can be distributed. But EDGE makes a substantial addition to existing work on this matter, and this white paper summarizes it.

While the scientific method allows us to *describe* the expected distributional impacts and public perception, we make an effort to deduce what these findings may imply for policy-making. We do this in the *Implications* sections of the different themes and summarize in the concluding Section 6.

Employment	Employ	Employ			
Policy costs / revenues		(Ele)	(Ele)	(Ele)	Ele
Public opinion & Perception	Res	Res	Res	Res	Res
				PO&P	
	Industries / Consumers	Regions	Household composition	Owners / Tenants	Income levels

Figure 2: Dimensions covered by different EDGE studies. Parenthesized entries denote dimensions that are discussed in the respective publications but do not feature in this white paper.

2 Distribution of employment impacts of the transformation in the electricity sector

Introduction

The transition towards an energy system with a high share of renewables in Switzerland will have broad socio-economic implications. Short of assessing all economic impacts of the energy transition, EDGE aims at analysing the distributional impacts on employment, including differences across industries and regions in Switzerland.

The goal of the study [1] is to find out how the distribution of employment across industries and regions would evolve in different scenarios for the transformation of the Swiss electricity system. Based on results of the electricity system model EXPANSE [2, 3], techno-economic data are combined with input-output modeling to estimate employment by industries and by regions.

We consider four scenarios of the EXPANSE model: two domestic scenarios combined with two scenarios of the development of the European electricity system. Regarding the domestic electricity system, one scenario assumes ambitious renewable electricity targets in 2035 and 2050, a certain flexibility of electricity demand, and no new nuclear power plants in 2050 (short: RET scenario). The counterfactual scenario assumes new nuclear power plants in 2050, no renewable electricity target, and no demand flexibility (short: no-RET scenario).

Findings

The employment impacts are calculated as the average annual employment between 2021 and 2050, needed for constructing, operating, and dismantling power plants in the RET scenario minus that needed in

the no-RET scenario. Employment impacts by region or by industry are related to the current total employment in the respective region or industry to normalise the results. We average the results for the two European scenarios for the sake of brevity.

Our results show that employment impacts are higher in the RET scenario than in the no-RET scenario. The main reason for this result is that more electricity is generated domestically in the RET scenario and less electricity is imported. Furthermore, with a high share of renewables, more capacity must be installed for sufficient winter production, as their specific yield is smaller than that of nuclear power plants. More capacity implies more investment, operation and dismantling activities, and finally higher employment.

The employment impacts are unequally distributed across industries. Not surprisingly, they are the largest in the power plants industry. The construction industry, the rest of the industry sector, and "planning and engineering" experience above average employment impacts.

Turning to regional employment impacts, we first report results for cantons and then for three broad regions: cities, Alps (municipalities in the Alps excluding cities), and midlands (all other municipalities). The RET scenario leads to more employment than the no-RET scenario in all cantons, but unequally so, as shown in Figure 3. The regional distribution of employment impacts depends strongly on how the utilisation of electricity generation technologies changes between the scenarios, where additional power plants are located, and how the regions participate in the supply chains of constructing, operating, and

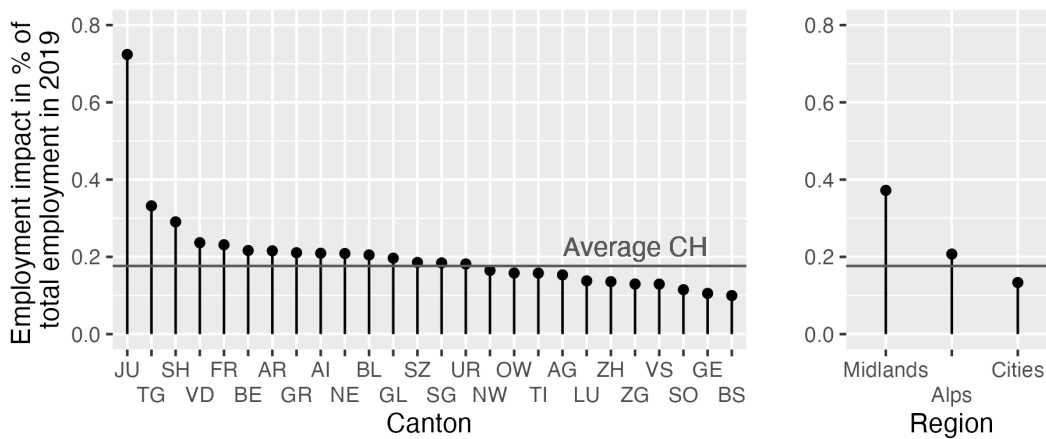


Figure 3: Employment impact by canton and region

dismantling power plants.

We also analyse whether the employment impacts contribute to decreasing the economic inequalities between cantons. We consider several indicators for the economic performance of cantons with specific strengths and limitations: GDP per capita, GDP per employed person in full-time equivalents (FTE), the resource potential per capita (used in Switzerland as a basis for inter-cantonal fiscal equalization), and the unemployment rate. The employment impacts across cantons are weakly negatively correlated with GDP per capita (correlation coefficient -0.23), GDP per employed person (-0.23) and the resource potential per capita (-0.30). Thus, economically weaker cantons tend to enjoy larger employment impacts. The correlation with the unemployment rate is near zero (0.09).

Turning to broad regions, our results show that employment impacts are largest in the midlands with approximately 0.4% of total employment, followed by the Alps (0.2%) and cities (0.13%).

Our findings depend on specific scenario assumptions and modeling results that should be taken into ac-

count before generalising the results.

Implications

Our results show that employment impacts are larger in all industries and regions with the investments and operating work needed for a scenario with ambitious renewable electricity targets and without new nuclear power plants, compared to a scenario without RET and with nuclear power plants in 2050. Employment impacts differ strongly between industries and between cantons and regions. They weakly contribute to decreasing economic inequalities between cantons.

These results help in identifying how sectors and regions may be affected by the transformation of the electricity sector. The employment potential generated by the transformation of the electricity system can only be exploited if the necessary labour force and skills are available. The results may also help in anticipating support and opposition from actors to energy system transformation policies. If policymakers wish to reduce inequalities between cantons or regions in the context of the energy transition, subsidies for renewable energy deployment could be targeted to underprivileged regions.

3 Distributional effects of promoting electricity demand flexibility through dynamic tariffs

Introduction

Dynamic tariffs are an effective way to ensure efficient deployment of heating and electricity-generating technologies. They can give incentives to households to behave in a system-friendly manner and help to keep in check the expansion of the electricity generation and grid system required by the energy transition. But dynamic tariffs are unpopular with consumers, because their household-specific impacts are hard to predict.

Research by EDGE set out to find patterns in household-specific outcomes under the introduction of dynamic tariffs [4]. The study analyzes how dynamic tariffs affect households of different population groups. It uses a rational agent model to determine, on the level of single buildings, what the optimal deployment and use of energy infrastructure is, when a pre-determined level of energy services (heating, operation of battery electric vehicles [BEVs], lighting, etc.) needs to be provided to inhabitants of the building. Electricity load patterns depend on the type of building (single-family or multifamily home) and on ownership of heat pumps or BEVs. Detailed data on building properties and geography give us information about the possibilities of residents to exploit PV combined with batteries, EVs, and heat pumps to optimize their own consumption. When tariff design changes, they react by adjusting their investments in solar PV and batteries, and they shift electricity demand from times of high tariffs to times when electricity is cheap.

Findings

Figure 4 shows how direct impacts on consumer welfare are distributed

across households of different income groups. (Consumer welfare refers to the individual benefits derived from the consumption of goods and services; it depends on both consumer prices and household income.) In the short run (see panel A), dynamic tariffs tend to yield losses, except for a small share of the population that is concentrated in the wealthiest income group (they already have the possibility to react to price signals by investing in solar PV and batteries). In the long run (panels B and C), all income levels are able to react to dynamic tariffs and profit from the switch since heat pumps and BEVs will be widespread. We find (see [4] for details) that additional benefits from installing PV and batteries accrue mostly to the higher-income groups since they are living in buildings that are well suited to these technologies. Reductions in energy costs for low-income groups come from deploying existing energy infrastructure (heat pumps and EVs) in a flexible manner. Low-income households and retirees are at a slight disadvantage compared to other groups in the short term, but this disadvantage will vanish in the long term provided that heat pumps and BEVs become pervasive. We find the heterogeneity in policy impacts to be considerable across households within all subgroups of the Swiss population.

A factor that the study was not able to consider is that additional regressivity may be introduced when grid tariffs have to be increased. This may well happen when the grid has to be financed through grid charge revenues from fewer consumed kWh (since dynamic tariffs incentivize own production with solar PV and batteries).

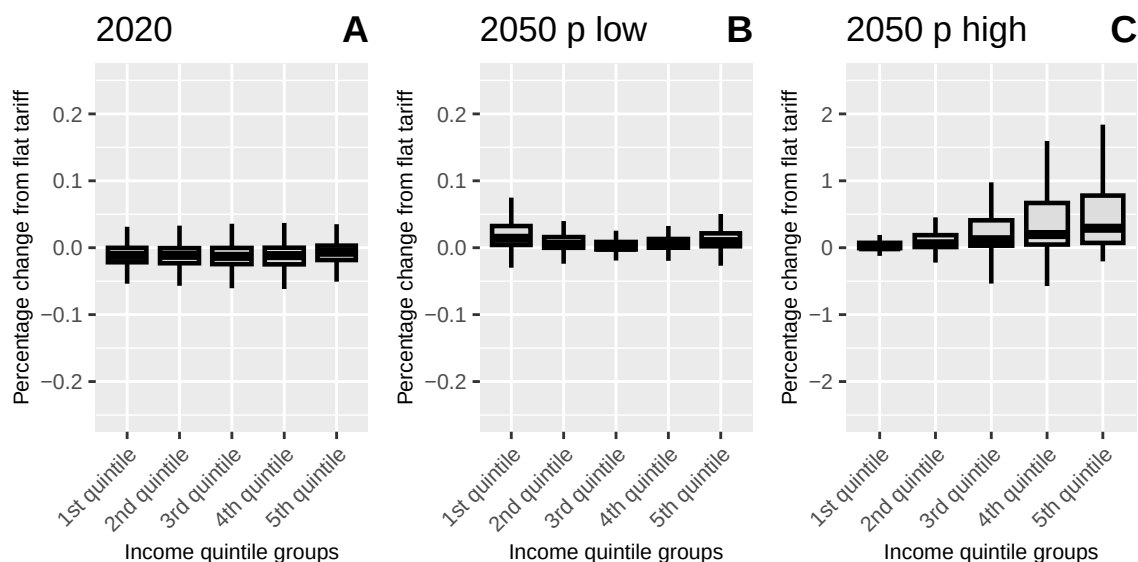


Figure 4: Direct impacts on household consumer welfare and their distribution across income levels for three scenarios. For 2020 (panel A), price levels, heat pump installations, and BEV ownership are according to historic data. For 2050, all cars are BEVs, heat pumps are installed where they are the most viable renewable energy option, and electricity price levels are low (panel B) or high (panel C) according to two different assumptions about transmission capacities to the neighbouring countries.

Implications

While no strong and persistent trends differentiate the household-level impacts across population groups, the heterogeneity of impacts within population groups is large. Industry experience and studies conducted outside EDGE (and outside of Switzerland) show that the uncertainty entailed by this heterogeneity is unnerving for consumers and makes dynamic tariffs unattractive. Offering opt-out options to dynamic tariffs (as is currently mandated by federal legislation) makes dynamic tariffs politically more palatable. Other options include designing consumer group-specific deals for transitioning to dynamic tariffs, but in order to reach equitable outcomes, the design of such group-specific deals (e.g., the specification of these groups) may require regulation.

To ensure that households can fully exploit intra-day price differences in the electricity system, a fast roll-out

of heat pumps, BEVs, and smart devices for controlling them is helpful. While utility companies can, and do, make dynamic tariffs available to their customers, there may be a role for strengthening and speeding up the introduction of regulation that ensures that households and building owners have technologies available to them to adapt to flexible prices (see, e.g., module 14 of the MuKEN 2025). In order to react optimally to dynamic electricity tariffs, renters of their dwelling have to have BEV charging infrastructure available and should be in control of their flexibility, and they would need their landlords to install efficient and flexible heating systems. Low-income households have to be able to afford heat pumps and BEVs. If upfront costs are prohibitive, “energy-as-a-service” or “rent-to-buy” business models can be a way to offer credit-constrained households affordable energy. These offers have to be available, and households need to be informed about their options.

4 Distributional effects of the defossilization in the residential sector

Introduction

Direct fuel combustion of the Swiss residential sector still accounted for 20% of total energy-related CO₂-eq emissions in 2023 [5], which underscores the need for more stringent defossilization policy mixes. The challenge, however, is not only to identify which instruments can bring the sector closer to its climate targets and at what costs, but also to understand whether adverse burdens across households and other actors are at risk, and how these can be mitigated. This is particularly important in the current policy context, marked by debates on the restructuring of building subsidy programmes [6, 7], differences in cantonal approaches [8], and the new so-called MuKEN 2025 model regulations in the energy sector [9]. Results presented here synthesize findings from two studies based on microsimulation using survey data [10, 11].

Findings

Policy mixes combining MuKEN regulation (bans on installing fossil and electric heating boilers, obligations to install PV panels when renovating roofs, and obligations to retrofit the worst insulated dwellings) with varying price-based incentives could promote 35% to 68% additional emissions savings beyond those that could be achieved through attractive energy investments on purely economic grounds without any incentive [10]. At the same time, these policy mixes would not affect all households equally. On average, less affluent households, older single-person households, homeowners, rural households, and households in less densely populated cantons would face the least favorable

investments and thus risk bearing the greatest direct cost burden [10]. Relative to disposable income, these groups would require higher upfront investments and would be less able to achieve low heating and electricity costs. Including an obligation for deep retrofits risks placing a particularly high burden on older men living alone, as they more often live in larger and worse-insulated houses, have relatively low disposable income, and high energy consumption.

Policy mixes that go beyond the MuKEN recommendations with stricter regulations can avoid even more greenhouse gas emissions (Figure 5), as illustrated by the difference in Figure 5 between the more ambitious policy mixes (the dotted and dashed lines) and the baseline mix with MuKEN regulations (the solid line). Yet, they still risk missing sectoral targets [11]. This applies to mixes that additionally include a ban on the use of fossil fuel heating by 2035, an obligation to install PV on all roofs by 2035, and an obligation to retrofit a wider range of buildings by 2035, which reflect forefront policies discussed in some cantons [12, 13]. Depending on subsidies and protection measures in the mix, these stricter regulations also risk generating unintended side effects. In the short term, they may create liquidity constraints for some dwelling owners because of high required investments, especially for retrofits. They may also increase pressure on public spending if subsidy levels are expanded (Figure 5). Even though they could benefit the construction industry by providing a peak of revenue in the next decade, they could lead to workforce bottlenecks. In the longer term, stricter regulations that sharply reduce fossil fuel and elec-

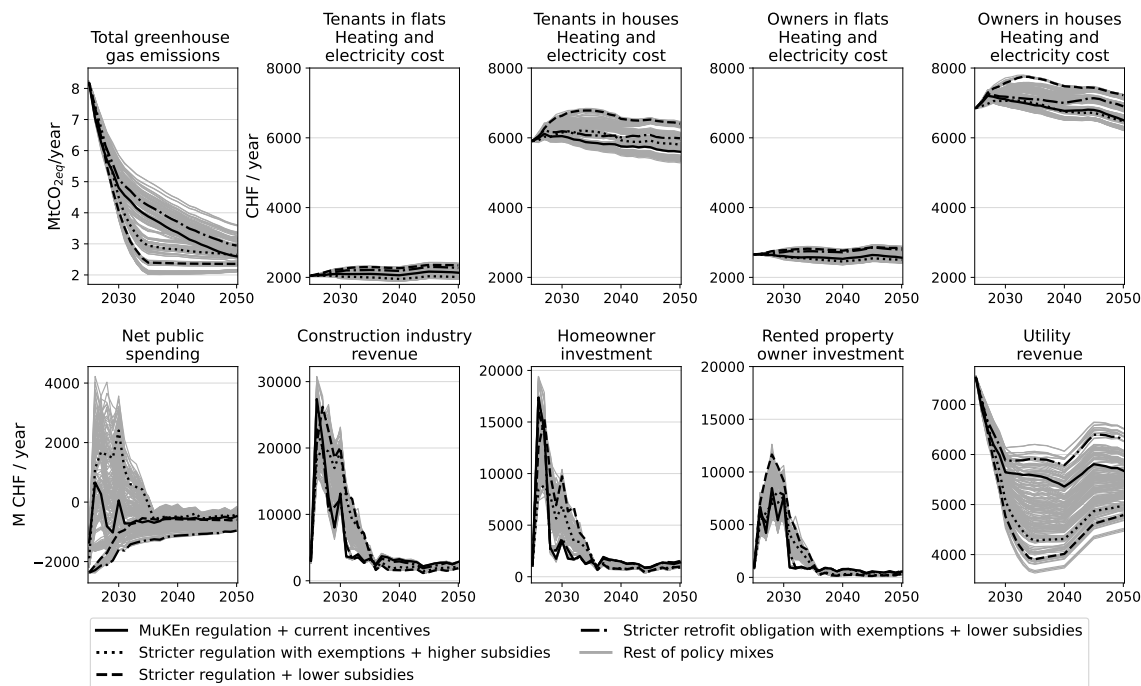


Figure 5: Defossilization pathways and key distributional outcomes of 192 policy mixes to defossilize the Swiss residential sector. Policy mixes vary in the stringency of heating bans, PV and retrofit obligations, subsidy levels, and protection measures for households at risk. Household costs include annualized investments.

tricity demand may reduce revenues for utilities while increasing the absolute energy-related costs for certain households, particularly those living in houses.

Implications

To minimize adverse effects, enforcement waves for regulation, incentive schemes, and protection measures should be carefully co-designed. Five implications follow:

1. Target financial support carefully. Especially with stricter regulations, subsidies and other support instruments should be well targeted to reduce the burden on vulnerable dwelling owners while avoiding unnecessary cross-subsidization and public spending.

2. Use exemptions sparingly. They may be justified in cases of hardship, but should remain limited to avoid undermining policy effectiveness. Other measures, such as low-interest loans,

limits on the pass-through of investment costs from rental property owners to tenants, and complementary social policies, could also be considered.

3. Anticipate adjustment needs for utilities. As building energy demand and fossil fuel use decline, utilities may face declining revenues and shifting service demands. Policymakers should anticipate these pressures and support the adaptation of utilities' business models.

4. Coordinate regulatory ambition with delivery capacity. Regulations should be phased in at a pace that matches the availability of skilled labour, while they should be expanded to meet the demands of the transition.

5. Complement pragmatic MuKEN-like regulations that trigger investments with policies to reduce embodied emissions in new required investments and energy use.

5 Public opinion and perceptions of distributional effects in the residential sector

Introduction

In the current times of poly-crises, including the climate, housing, and cost-of-living crises, defossilization of the residential building sector is urgent but is likely to generate new social risks and conflicts over the distribution of costs. To understand how the public views the distributional consequences of policies to reduce GHG emissions in the residential building sector, we move beyond previous research, which usually focused on market-based policy measures (such as carbon taxation), and study regulatory climate policies (see [14]). Thus, we performed an experiment in the second SWEET EDGE survey, a probability-based sample of Swiss residents fielded between May 10 and August 4, 2025. All respondents were informed (in random order) about two policy measures and their (equal) effectiveness in terms of GHG emissions reductions inspired by the *MuKE*: a ban on installing fossil-fuelled heating systems and an obligation to retrofit building envelopes of the worst insulated buildings [15]. One experimental group also received information on the distributional impacts (*Distrib. Info.*) for both tenants and homeowners, i.e., how many would pay more or less as a result of the policy. In a second experimental group, the policy was enhanced with an element of social protection for vulnerable households (either policy exemptions that reduce effectiveness by 5 percentage points or targeted subsidies, which keep policy effectiveness constant). For this combined policy (*with Redistribution.*), the respective distributional impacts were displayed to respondents in the same style as in the other experimental group.

Findings

Our analysis, presented in Figure 6, highlights that the moment the distributional costs of residential defossilization policies become visible, they strongly shape public opinion. Protection measures that target only financially vulnerable households thereby fail to recover the drop in policy support by making distributional impacts visible.

Homeowners and tenants may be affected quite differently by these policies. While the former would be urged to invest in new heating systems or retrofitting, tenants may fear higher rents. The results, however, show that the reactions to the experimental treatments are very similar across groups.

Implications

Our study reveals that visible distributional costs heavily affect public opinion about regulative policy measures to accelerate the energy transition. This aligns with findings from other research on market-based policy instruments, such as carbon taxes. Moreover, the protection measures we tested – targeted exceptions and additional subsidies for financially vulnerable households – failed to increase public support, likely because they did not create a clear perception of improved distributional impacts.

Our findings, therefore, imply that public support for large-scale implementation of coercive policies, such as retrofit obligations or fossil fuel heating bans, is likely to decrease when distributional impacts enter public discussion.

A second implication is that, to increase public support, one might

need stronger protection policies and/or better communicate how protection measures improve distributional outcomes. We expect compensation measures to be effective if they create a sense that vulnerable groups are better protected, that impacts of the distribution are more equally distributed, that outcomes are generally fairer, or that individuals feel protected or better off themselves, given that self-interest also plays a role.

Moreover, these measures likely need to compensate “vulnerable groups” more broadly. While renting is most common in Switzerland, meaning that protection measures will be particularly relevant to win political majorities, compensation likely

also needs to include homeowners who bear the direct costs of regulatory policies in the building sector. A concrete policy option to protect financially vulnerable tenants would be to limit and control the pass-through of the costs of regulation-induced retrofitting (i.e., by controlling rent increases and empowering tenants to fight them). For financially vulnerable homeowners, in addition to exemptions and higher targeted subsidies, pay-as-you-save financing mechanisms and zero- or low-interest loans could be considered. In future research, such policy configurations could be included in new large-scale population surveys to elicit public support for them.

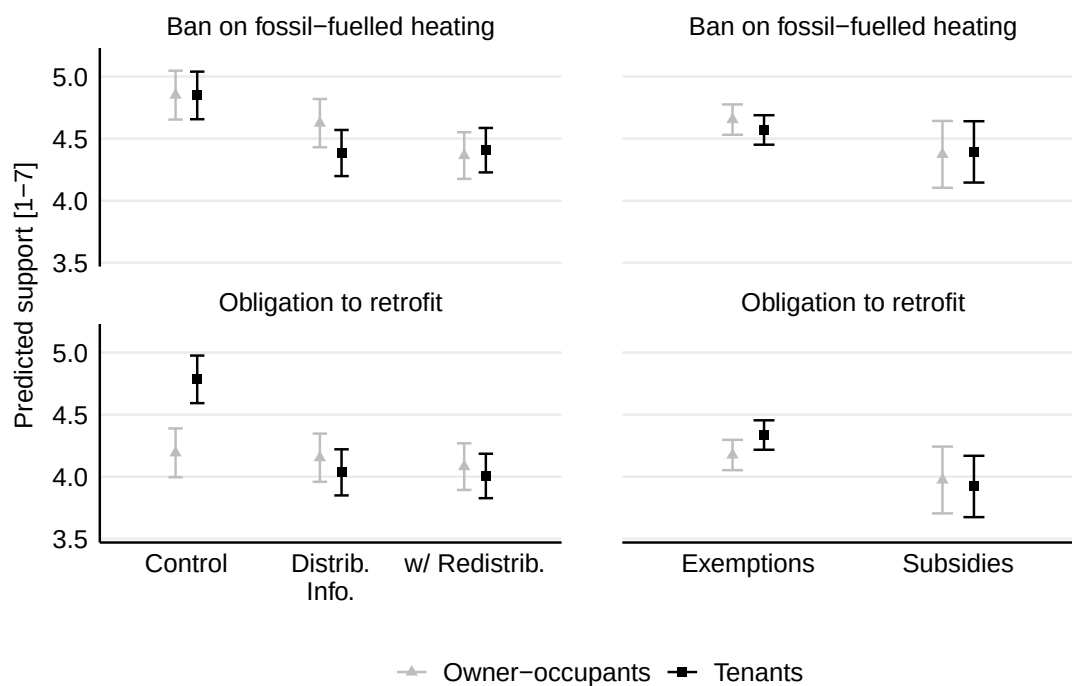


Figure 6: Predictions for policy support [measured on a 7-point Likert-scale, from “completely oppose” (1) to “neither support nor oppose” (4) to “completely support” (7)] based on a linear random intercept model using $n=1831$ respondents (950 tenants and 881 homeowners), split by policy. The left panel averages over the two redistributive elements, while the right panel displays the type of redistributive element.

6 Conclusion

The research activities of the SWEET consortium EDGE yield a broad set of insights regarding the distribution of impacts of the Swiss energy transition.

When studying the employment impacts of implementing renewable electricity targets, EDGE's research finds that employment linked to constructing, operating, and dismantling power plants will increase under renewable electricity targets. Employment is higher in all cantons and regions, but the level of employment impacts tends to decrease economic inequalities between cantons by boosting employment, particularly in cantons with lower GDP.

A second strand of research analyses how dynamic electricity tariffs, designed to increase the efficiency of the electricity system, impact different parts of the Swiss population. The findings indicate that the distribution of impacts of introducing dynamic tariffs does not give cause for grave concern about burdening certain population groups with disproportionate policy costs; rather, it shows that, within all subgroups of the Swiss population, the heterogeneity of impacts is large. The fact that some households may be less well off under dynamic tariffs than without them may be unpopular with consumers/voters, and opt-out options can help avoid both the negative impacts of the introduction of dynamic tariffs and disapproval by consumers.

Third, EDGE's research examines the effectiveness and distributional implications of a range of policy mixes aimed at defossilizing the Swiss residential sector. The studies find that, on average, less affluent households, older single-person households, homeowners, rural households, and households in

less densely populated cantons risk bearing a greater direct cost burden under policy mixes combining proposed MuKE regulation with price-based incentives. Similar but more stringent policy mixes forcing investments by 2035 would accelerate defossilization, but depending on the complementary incentives and protection measures, they may have unintended side effects such as liquidity constraints for some dwelling owners, increased pressure on public spending, workforce bottlenecks, reduced revenues for utilities, and increasing energy-related costs for certain households.

Survey results indicate that visible distributional effects of ambitious defossilization policies have the potential to significantly reduce their public support. Moreover, targeted exemptions and additional subsidies for financially vulnerable households were unable to increase public opinion, likely because they did not create a clear perception of improved distributional impacts.

In summary, the following general recommendations can be made to policymakers.

1. Make sure that the Swiss economy is well prepared for the changes entailed by the defossilization of the Swiss economy. Re-employment and re-skilling of workers may help the Swiss energy transition by providing a sufficient workforce to build up the decentralized, renewable energy infrastructure.
2. Ensure that all actors have the right incentives and information. Split incentives and inattention in the owner-tenant relationship may lead to investments in demand flexibility that would be desirable for tenants not being made.

3. Credit constraints can restrict consumers' access to efficient and flexible technologies. Low-interest loans, "energy-as-a-service", and "rent-to-buy" business models can help. Offers should be available to and known by consumers.

4. Transparent and proactive communication may be key. If policy impacts are heterogeneous, different political actors can select different examples to promote their views on the policy. The resulting contradictory arguments can leave the impression that policy options are poorly understood or not well thought through.

5. Heterogeneity in impacts violates

the equal sacrifice principle and may appear unattractive to the population and thus to voters. For policy proposals to succeed, the benefits should be emphasized, and the arguments that the benefits outweigh the possible inequality in and uncertainty about the distribution of policy impacts should be convincing.

6. To the extent that coercive policies risk being too burdensome in certain cases, well-targeted complementary policy instruments are recommended to compensate "vulnerable groups" more broadly. In particular, they should protect not only tenants but also homeowners.

7 Acknowledgements

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SWEET EDGE

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